Garuda: Privacy-First Edge-AI Home Security

Brief project presentation based on the paper draft

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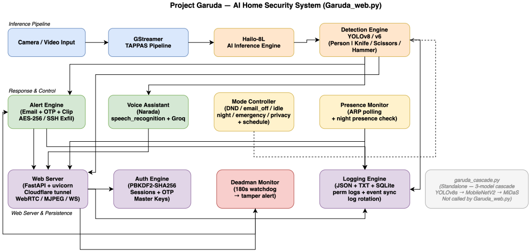
Problem and Goal

- Cloud cameras are convenient, but they send private footage off-device and usually require subscriptions.
- DIY home-security builds keep data local, but many stop at simple motion detection and weak alert pipelines.
- Garuda targets the gap: accurate threat detection, local processing, secure access, and practical deployment on low-power hardware.
- The paper focuses on a measured system integration result, not a new detection algorithm.

Core Idea

- All camera inference stays on-device.
- Raspberry Pi 5 hosts the services layer.
- Hailo-8L NPU handles real-time neural inference.
- FastAPI dashboard, remote access, alerts, logging, and voice control run alongside detection.

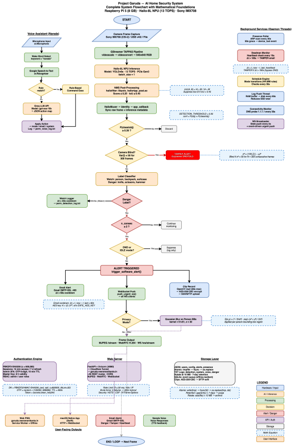


Design priority: privacy first, real-time response, low recurring cost.

System Pipeline

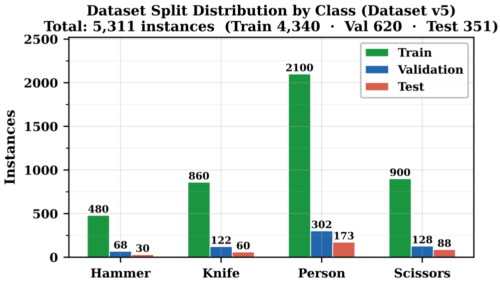
1. Camera frames enter a GStreamer + Hailo TAPPAS pipeline.
2. YOLOv8s performs primary object detection.
3. Danger labels are verified through a secondary crop classifier.
4. Alerts, dashboard updates, and encrypted evidence handling run asynchronously.

- Bounded queues prevent slow side effects from blocking inference.
- CPU affinity keeps capture, inference, verification, and services separated.



Models and Data

- Primary detector: fine-tuned YOLOv8s for 4 classes.
- Classes: Person, Knife, Hammer, Scissors.
- Secondary verifier: MobileNetV2 on cropped detections.
- Extra pre-filter: MiDaS Small depth-variance cue for person-related screening.

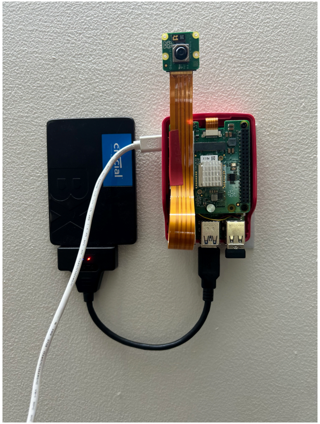


Dataset snapshot

Final v5 corpus: 6,226 images, 5,311 annotated instances, 80/10/10 split.

Hardware and Deployment

- Host: Raspberry Pi 5 with active cooling.
- Accelerator: Hailo-8L AI HAT+ (13 TOPS, INT8).
- Camera: Raspberry Pi Camera Module v3.
- Storage: SSD for models, logs, clips, and database.

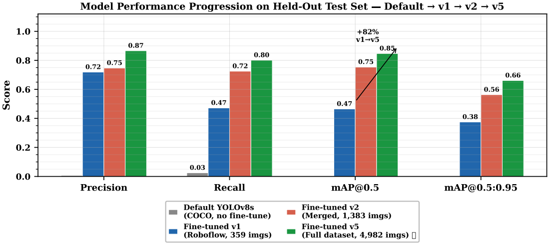


Deployment principle

Neural inference uses the NPU; web, voice, alerting, and logging stay on the CPU side.

Key Detection Results

- Test mAP@0.5 improved from 0.465 (v1) to 0.847 (v5).
- Overall test precision: 0.866.
- Overall test recall: 0.801.
- Best class: Scissors with mAP@0.5 of 0.967.

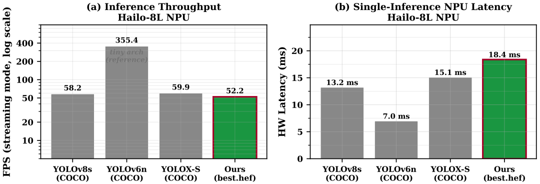


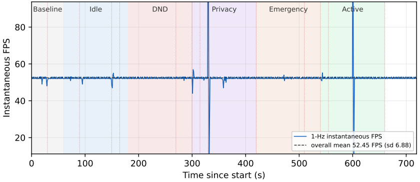
Interpretation

Performance gains mainly came from better dataset scale and diversity, not from changing the detector family.

Runtime Performance

- On-chip throughput: 52.2 FPS.
- On-chip latency: 18.4 ms.
- Mean FPS across operating modes: 52.26 ± 0.76.
- Board power under load: 5.89 ± 1.43 W.





Why this matters

The pipeline stayed stable even while alerts, clip uploads, voice queries, and streaming were active.

Privacy and Security Features

- No image or video frames are sent to cloud inference services.
- Remote access is exposed through a Cloudflare tunnel instead of open port forwarding.
- Evidence clips are encrypted with AES-256-CBC before upload.
- Authentication uses PBKDF2-SHA256 with one-time-password support.
- Privacy mode can blur person detections before any on-LAN viewing path.
- Voice fallback sends short text only; raw audio and video remain local.

Main Takeaways

- Garuda shows that a practical, privacy-first home-security stack can run fully at the edge on commodity hardware.
- The strongest contribution is system engineering: model integration, asynchronous design, and measured deployment behavior.
- The platform combines detection quality, real-time speed, and secure service integration without recurring cloud dependence.

Next steps

Longer real-world deployment studies, nuisance-condition testing, stronger anti-spoofing, and broader class coverage.

Thank you

Questions?